

# Titan’s Equatorial Dunes Broadly Follow Lambertian Reflectance

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## ABSTRACT

**NOTE: Red notes are important! Do not submit the document with any of them remaining!**

The nature of Titan’s surface is poorly understood, largely due to the atmosphere’s extensive interference. To handle this interference, we simulate Lambertian Titan models using the radiative transfer program `SRTC++` at different surface albedos. We then compare these models to real data from Cassini VIMS (Visual and Infrared Mapping Spectrometer) of the Huygens landing site and equatorial dunes. We confirm that `SRTC++` produces reasonable simulations for Lambertian surfaces shrouded by Titan’s atmosphere in many, but not all, situations. Furthermore, we find that Titan’s equatorial dunes behave as Lambertian surfaces, with all identified deviations correlating to suspected atmospheric model deficiencies. We retrieve the surface albedos of Titan’s dunes in the eight IR (infrared) atmospheric windows and find the albedos largely consistent with multiple prior works, but inconsistent with other prior works, revealing a disagreement over what the surface albedos of Titan’s dunes truly are. Lastly, we perform a targeted search for an opposition effect on Titan’s dunes through VIMS observations and find none, though the narrowness of the effect may make it unobservable.

*Keywords:* KEYWORDS (111) — KEYWORDS (112)

## 1. INTRODUCTION

Titan has one of the least understood surfaces in the entire Solar System, due largely to its thick haze-filled atmosphere that is opaque to most light. While there do exist a handful of atmospheric “windows” through which specific wavelengths of light can pass relatively unimpeded (Barnes et al. 2007), they permit us to glean only limited spectral information on the surface. Even within the windows, the thick atmosphere contaminates the relatively small amount of surface information we do receive. Transmission is unavoidably a combination of surface and atmospheric effects (Es-sayeh et al. 2023). To separate surface and atmosphere contributions, we turn to radiative transfer models of Titan. These models calculate how Titan’s atmosphere alters light passing through it, including any effects the surface might have as light interacts with it on the journey to a modeled detector. These models allow for true surface effect identification by comparing real data to the simulations. Many groups have created radiative transfer models over the years, each with their own strengths and weaknesses (Vinatier et al. (2007); Griffith et al. (2012); Xu et al. (2013); Barnes et al. (2018b); Rodriguez et al. (2018); Corlies et al. (2021); Rannou et al. (2021); and Es-sayeh et al. (2023); among others). These radiative transfer models depend on accurate knowledge of Titan’s atmosphere. We have the best measurements at the

moon’s equatorial regions where the Huygens lander sampled the atmosphere in situ (Tomasko et al. 2008).

Many groups have performed surface characterization studies attempting to filter out the influence of the atmosphere in the past (McCord et al. 2006; Buratti et al. 2006; Soderblom et al. 2009; Kazeminejad et al. 2011; Bonnefoy et al. 2016; Brossier et al. 2018; Es-sayeh et al. 2023; Solomonidou et al. 2024). However, the majority of studies assume that the surface behaves as a Lambertian reflector, an isotropic scatterer with no directly reflected or backscattered components. Buratti et al. (2006) is a notable exception.

Few, if any, surfaces in the Solar System exhibit truly Lambertian behavior, largely due to the prevalent opposition effect (Déau et al. 2009; Schröder & Keller 2009). We might expect Titan’s surface to exhibit similar non-Lambertian behaviors, but first-order analysis of the moon as a whole in the IR (infrared) shows a Lambertian surface (Le Mouélic et al. 2019, 2012).

On the other hand, the one observation we have from the ground on Titan, the Huygens lander, did report an opposition effect (Schröder & Keller 2009; Karkoschka et al. 2012). However, Huygens landed in a “dark blue” region (Rodriguez et al. 2006), an uncommon terrain on Titan’s surface (Keller et al. 2008). For Titan as a whole, Radar observations show an opposition effect (Neish et al. 2010; Wye 2011), but given the drastically different length scales probed between radar

(centimeters) and IR (microns), we have no reason to expect similar reflectance behavior. Furthermore, other ways to deviate from Lambertian behavior exist, such as a forward-scattering component to the surface phase function, which may result from wetted terrains, for example (Dhingra et al. 2020).

Our primary science goal within this paper is to better understand Titan’s surface properties by creating Lambertian Titan radiative transfer simulations and comparing them to real observations of the moon’s equatorial dunes. We accomplish our goal through identifying the degree of Lambertian behavior from Titan’s dunes in reflected IR sunlight. In the process of this Lambertian analysis, we gather information about the dunes’ albedo in various IR wavelength windows. We also identify differing behaviors between said windows.

The highest-quality observations of Titan’s surface come from the Cassini mission, observations which have a wide spread of viewing geometries. Having such a variety of observations helps our characterization of Titan’s surface, making it easier to determine how non-Lambertian a terrain is. Most current radiative transfer models applicable to Titan either assume a plane parallel atmosphere in their calculations (Griffith et al. 2012; Rodriguez et al. 2018; Es-sayeh et al. 2023), don’t consider angle at all (Vinatier et al. 2007; Rannou et al. 2021), or use a spherical approximation (Corlies et al. 2021). Thus, all these radiative transfer models lose accuracy at high incidence and/or emission angles, and would miss potential non-Lambertian effects.

To gain the useful information contained within observations at oblique viewing geometries, we must consider the spherical nature of Titan’s atmosphere. Thus, we create forward-modeled simulations of a Lambertian Titan using SRTC++ (Spherical Radiative Transfer in C++), a radiative transfer code tailored to model Titan in full spherical geometry at the infrared wavelengths available to Cassini’s VIMS (Visual and Infrared Mapping Spectrometer) instrument (Barnes et al. 2018b). Other spherical models exist (Xu et al. 2013), but we choose SRTC++ due to familiarity with the code.

Now equipped with a spherical radiative transfer model and atmospheric characterization from Huygens, we can compare simulations with reality on a scale covering the entire Cassini mission. As the equatorial regions have the best characterized atmosphere, we choose to examine that terrain. While we examine multiple terrain types, we determine that the equatorial dunes have the highest number of reliable observations from which to draw meaningful conclusions.

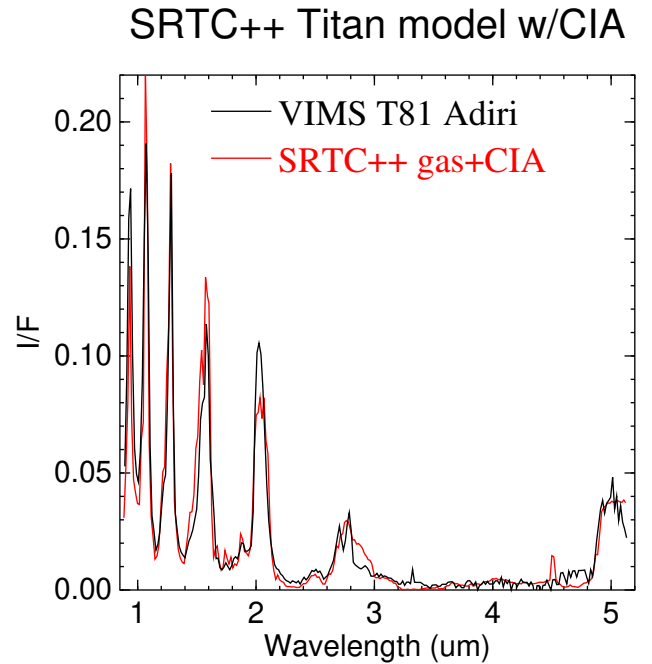
We also hand-pick observations from the Huygens landing site for the sake of comparison, even though that location was only observed at a limited number of viewing geometries. For both the dunes and the Huygens landing site, we compile VIMS observations from across the entire Cassini mission

over all available viewing geometries, limiting observations to only those we judged to have sufficient quality.

In addition to uncovering Titan’s surface properties, this work serves a secondary purpose—to validate the SRTC++ simulation against real data. We validate in two ways: qualitatively comparing the behavior of the dunes with the behavior of the simulations, and retrieving surface albedo to quantitatively check against previous groups’ estimates.

In this paper, we first outline improvements made to SRTC++ in Section 2 and report on the results of those changes in Section 3. We describe our Titan data in Section 4 and compare observation to simulation in Section 5 for both the Huygens landing site and equatorial dunes, while also retrieving surface albedos for the dunes and checking them against prior work. Lastly, we conclude in Section 6.

## 2. RADIATIVE TRANSFER MODEL METHODS



**Figure 1.** Here we compare a SRTC++ model Titan spectrum (red) to an observed spectrum (black) acquired by Cassini VIMS on the T81 flyby (2012 January 30) of Adiri, a bright highlands region near the Huygens and Dragonfly landing sites. The incorporation of correlated-k coefficients to model atmospheric absorption follows Es-sayeh et al. (2023), including the model atmosphere and surface albedo inputs. We did not tune the surface albedos and atmospheric parameters to fit the observation spectrum; rather, the intent of this plot is more to show the strong general agreement between observations and the model instead of reproducing this particular spectrum.

Spherical Radiative Transfer in C++, or SRTC++, is a fully spherical Monte-Carlo radiative transfer model designed to address problems requiring spatial variability in properties and/or high incidence and emission angles (Barnes et al.

141 2018b). It computes in parallel the propagation, gaseous ab-  
 142 sorption, and aerosol scattering for billions of photon pack-  
 143 ets to build up synthetic images of Titan at designated wave-  
 144 lengths. The strength of SRTC++ lies in its ability to treat  
 145 spatially variagated atmospheres in cases where extreme ge-  
 146 ometries affect output and to solve spatial problems moreso  
 147 than spectral problems. The SRTC++ algorithm has previ-  
 148 ously simulated trans-terminator surface illumination on Ti-  
 149 tan (Barnes et al. 2018a), the effect of atmospheric scatter-  
 150 ing on surface imaging from Titan orbiters (Barnes et al.  
 151 2020), and explorations of Titan’s adjacency effect (MacKen-  
 152 zie et al. 2025).

153 We upgraded SRTC++’s capabilities by adding gaseous  
 154 absorption, thereby allowing for it to model complete spec-  
 155 tra. We model the absorption using the correlated-k approach  
 156 outlined by Es-sayeh et al. (2023) by an explicit numerical in-  
 157 tegral of extinctions in four Gauss orders along the flight path  
 158 of each photon packet. The resulting opacity adds to that cal-  
 159 culated from haze scattering on each segment traversed.

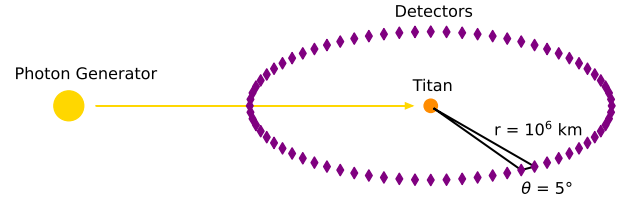
160 We show an example of SRTC++ results in this new spec-  
 161 tral mode in Figure 1. Figure 1 compares a generic forward-  
 162 modeled Titan spectrum from SRTC++ using standard Es-  
 163 sayeh et al. (2023) inputs to a typical observed Titan spec-  
 164 trum from Cassini VIMS on the T81 flyby. In general the two  
 165 spectra show strong agreement, although with some discrep-  
 166 ancies at the longward end of atmospheric windows (“known  
 167 unknown” absorbers, Barnes et al. (2026)) and in the shortest  
 168 wavelength windows where the effects of Rayleigh scattering  
 169 (as of yet unmodeled by SRTC++) come into play.

170 For our work in this paper, we use SRTC++ to create three  
 171 different simulations of a uniformly Lambertian Titan, each  
 172 run with a different surface albedo: 0.0, 0.1, and 0.2. Be-  
 173 sides the albedo, the simulations have identical parameters,  
 174 as shown in Figure 2. The virtual detectors sit 10,000 km  
 175 away from the Lambertian Titan, separated by 5 degrees and  
 176 completely encircle the moon. The simulation outputs at all  
 177 eight VIMS IR wavelength windows for Titan’s atmosphere:  
 178 0.93, 1.08, 1.27, 1.59, 2.01, 2.69, 2.79, and 5.00  $\mu\text{m}$ .

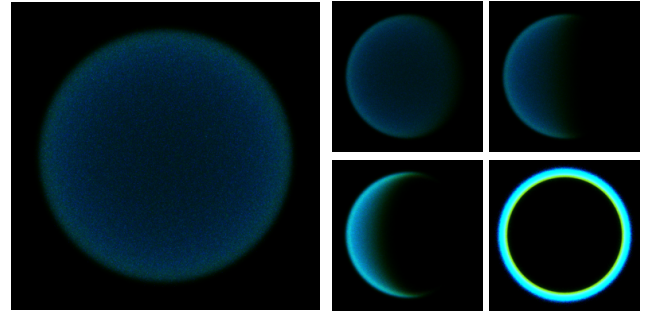
### 179 3. SIMULATION RESULTS

180 Figures 3 and 4 show images from simulations at 0.0 and  
 181 0.1 surface albedo, respectively, colored with 5.00  $\mu\text{m}$  as red,  
 182 2.01  $\mu\text{m}$  as green, and 1.27  $\mu\text{m}$  as blue, as done in Barnes  
 183 et al. (2005). The results of 0.1 albedo resemble the “green-  
 184 ish” pictures of Titan in Barnes et al. (2005), though natu-  
 185 rally without any terrain variation. Not shown is a simulation  
 186 run at 0.2 albedo, a result similar to the 0.1 simulation, but a  
 187 brighter green.

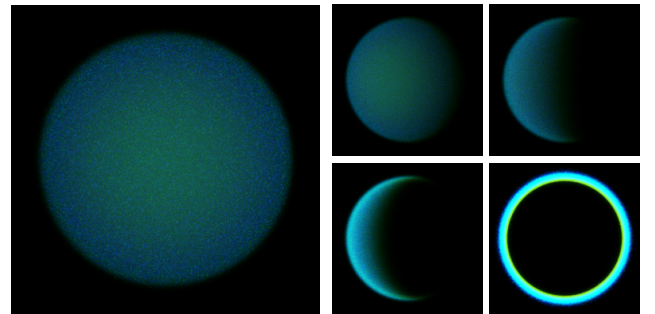
188 The simulations generally follow expectation for a Lam-  
 189 bertian sphere obscured by a thick atmosphere. Pont & Koen-  
 190 derink (2007) show what a Lambertian sphere should look  
 191 like without an atmosphere in their Figure 9, and our 0.0



**Figure 2.** Layout of our SRTC++ simulations. All distances are not to scale. The detectors are all equidistant from Titan with identical angular separation between them. The yellow arrow represents the path “photon packets” take when shot at Titan. The “photon packets” don’t interact with the detector they initially pass through on their way to Titan.



**Figure 3.** SRTC++ Simulation results for a Lambertian Titan with 0.0 surface albedo, colored with 5.00, 2.01, and 1.27  $\mu\text{m}$  mapped to red, green, and blue, respectively, as done in Barnes et al. (2005). The left image is viewed at 0° phase. The right four images are, clockwise from the top left, at 35°, 90°, 180°, and 120°. **Animating Version: A0.0LambertSimMOVIE.gif**



**Figure 4.** SRTC++ Simulation results for a Lambertian Titan with 0.1 surface albedo, colored with 5.00, 2.01, and 1.27  $\mu\text{m}$  mapped to red, green, and blue, respectively, as done in Barnes et al. (2005). The left image is viewed at 0° phase. The right four images are, clockwise from the top left, at 35°, 90°, 180°, and 120°. **Animating Version: A0.1LambertSimMOVIE.gif**

192 albedo simulation shows us the effect of the atmosphere with-  
 193 out any signal coming from the surface. We expect both at-  
 194 mospheric and surface effects in the 0.1 albedo simulation,  
 195 which we do see in the zero phase view: a flat greenish center  
 196 with minimal atmospheric influence that slowly fades off  
 197 as the emission angle becomes steeper, at which point the  
 198 atmosphere takes over with its bluish hue.

199 Haze scattering dominates the overall coloration; atmo-  
 200 spheric haze scatters bluer light more readily than redder  
 201 light. In the raw data, the blue channel exhibits the highest  
 202 brightness even in the center at 0.1 albedo. The chosen color  
 203 balancing gives Titan its greenish hue, a sensible choice as  
 204 it allows us to readily identify places with more surface sig-  
 205 nal than atmospheric. The center of the disc, when viewed at  
 206 zero phase, is generally uniform with small gradients toward  
 207 the limb.

208 Views other than at zero phase also behave regularly; atmo-  
 209 spheric effects dominate the limb, making it bluer. Light for-  
 210 ward scatters off of Titan’s haze particles more often than it  
 211 backscatters (García Muñoz et al. 2017; Cooper et al. 2025),  
 212 so that the closer the Sun gets to behind Titan, the brighter the  
 213 limb becomes. A Lambertian surface without an atmosphere  
 214 would have no light coming from anywhere that was not di-  
 215 rectly illuminated, but we see a surprisingly large amount of  
 216 light from beyond the terminator due to atmospheric scatter-  
 217 ing, an effect also seen in real observations.

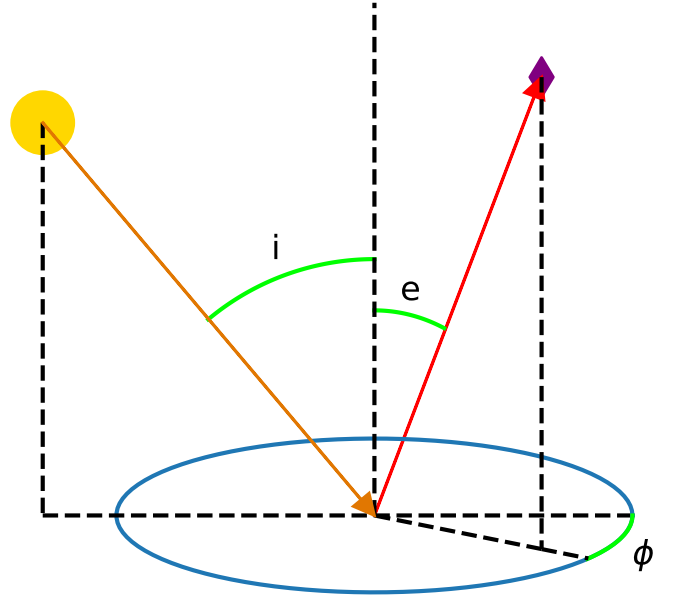
218 We can also see the shadow Titan casts on its own atmo-  
 219 sphere in the profile views by examining the abruptly dimin-  
 220 ished signal near the top and bottom of the moon, while the  
 221 atmosphere above this shadow remains brightly illuminated.  
 222 This effect happens on Earth as well; it causes the colors of  
 223 twilight (Lynch & Livingston 2004).

224 The simulation results appear slightly grainy due to the  
 225 Monte-Carlo nature of SRTC++. If we were to run the sim-  
 226 ulation for longer, the S/N (signal-to-noise ratio) would in-  
 227 crease, and the overall image result would become smoother,  
 228 albeit at the cost of computer time. The simulations that we  
 229 use represent a compromise between S/N and CPU-months.

230 We ultimately seek to compare these simulations with real  
 231 data. While we can qualitatively see that real Titan images  
 232 have similar coloration and limb effects (Barnes et al. 2005),  
 233 the fact that the real Titan has numerous different terrain  
 234 types interferes with any more robust conclusions. Thus, we  
 235 transform the simulations to present their data based on view-  
 236 ing geometry, rather than direct visual images. The trans-  
 237 formed simulations can, given a specific viewing geometry,  
 238 describe how a Lambertian Titan would appear at that geom-  
 239 etry. The simulations effectively transform into numerical  
 240 bidirectional reflectance distribution functions (BRDFs).

241 To construct these transformed simulations, we sort every  
 242 spectel into bins based on viewing geometry, and then we av-  
 243 erage their values. We separate the bins by  $5^\circ$  increments in

244 incidence, emission, and azimuth angle, defined as shown in  
 245 Figure 5, with  $0^\circ$  azimuth being the forward-scattering direc-  
 246 tion, and  $180^\circ$  being backscattering. This process increases  
 247 the S/N for the amalgamated spectels, ameliorating the shot  
 248 noise in the simulations.



**Figure 5.** Viewing geometry angles used in this paper. The incidence angle is represented by  $i$ , the emission angle by  $e$ , and the azimuth angle by  $\phi$ . The arrows trace out a path from the sun (yellow circle) to some arbitrary spot on Titan’s surface before and then to a detector (purple diamond). Diagrams modeled after this one appear in subsequent figures to give context for the data.

249 Notably, our transformed simulations use three angles,  
 250 while more traditional BRDFs have four. We do not treat  
 251 topographic changes in a surface due to the information’s  
 252 heavy dependence on context, so we can safely assume that  
 253 the Sunlight always comes in from the same relative direc-  
 254 tion, removing the need to keep track of the fourth view-  
 255 ing angle. As such, we will call these transformed simula-  
 256 tions restricted BRDFs from here on out. This choice also  
 257 makes it easier to determine when backscattering or forward-  
 258 scattering occurs, key indicators of non-Lambertian behavior.

259 We also differ from traditional BRDFs in another way: we  
 260 retain the atmospheric contributions. We produce the appar-  
 261 ent response Titan has to light when observed from outside  
 262 the atmosphere, not the direct response of the surface.

263 With 3 different albedos and 8 wavelengths, we have a to-  
 264 tal of 24 restricted BRDFs ready for comparison with other  
 265 restricted BRDFs compiled from real VIMS data.

#### 266 4. OBSERVATIONS

267 Cassini performed 127 close flybys of Titan during its mis-  
 268 sion (Seal & Bittner 2017), and most of those flybys have ob-  
 269 servations from Cassini VIMS. Orbital mechanics limits the

viewing geometries available on any single flyby. Thus, in order to gain a proper understanding of the surface of Titan at a wide range of viewing geometries, we use observations from as many flybys as possible.

We investigate the equatorial regions between 30°N and 30°S latitude, as this includes the Huygens landing site where Huygens’ directly sampled the atmosphere, making it the best characterized region on Titan (Tomasko et al. 2008). We also want terrain types that Cassini examined from a wide variety of viewing geometries with a large number of reliable observations. We judge that Titan’s equatorial dunes best meet all these criteria. To gather the relevant data, we create a raster mask of Titan’s equatorial dunes in Figure 6 at a resolution of one pixel per degree on Titan’s surface; 61 pixels high in latitude and 360 wide in longitude.

The creation of the mask begins with the Titan terrain map created by Lopes et al. (2020) using radar data. VIMS observations, taken in IR, often don’t match the radar observations (Soderblom et al. 2007), but tend to agree in the bulk of major features. Most notably (and importantly for this paper), the dunes have good agreement on a global scale (Soderblom et al. 2007; Radebaugh 2013; Rodriguez et al. 2014).

Our investigation procedure for the VIMS observations begins with a basic database search; in our specific case, we look for any images that have pixels in the equatorial regions between 30°N and 30°S latitude, where those pixels have 25 km ground resolution or finer. The database we use has already filtered out clearly erroneous observations, as well as any images that only had a handful of pixels in height. We calibrate the images with the standard VIMS pipeline (Barnes et al. 2007).

We put each pixel that matches our criteria into a list. We then bin spectral components of each listed pixel and create a restricted BRDF out of them in a process identical to that described in Section 3. Although we ran the simulations at precisely 5.00  $\mu\text{m}$ , for VIMS observations we average together sixteen separate channels, taking advantage of the large width of the 5  $\mu\text{m}$  window to increase the S/N. Unlike many other investigations (Buratti et al. 2006; Bonnefoy et al. 2016; Brossier et al. 2018; Es-sayeh et al. 2023), we do not attempt to subtract off or correct for atmospheric contributions, as our simulations include those effects.

The restricted BRDFs have a few limitations. Primarily, bins at certain viewing angles, usually at the extreme ends of allowed values, have no data since Cassini was never in those positions. Also, no checks for interfering clouds occur while amalgamating surface observations.

Coarse observations can mix spectral end members together while particularly fine-resolution images can separate them; for instance, the handful of VIMS observations that can see interdune areas (Barnes et al. 2008). Pure spectral endmembers need not match the behavior of the surrounding

terrain, and could conceivably offset the final model. Fortunately, even though the interdune areas vary considerably across Titan (Bonnefoy et al. 2016), the restricted BRDFs created for the dunes exhibit remarkable consistency and order.

We expect that any retrieved albedos from this model will most likely exhibit bias higher than pure dune sands, given the brighter interdunes (Bonnefoy et al. 2016). However, as we examine the dunes terrain type as a whole, we do not consider this bias a detriment to any of our results.

In addition to the dunes, we also derive restricted BRDFs for the Huygens landing site, allowing for more direct comparisons to observations made by the Huygens lander (Tomasko et al. 2008). We perform a database search similarly to how we make the dunes models, but instead of using the mask, we manually clean up any situations where we obtain the wrong latitude and longitude due to ephemeris uncertainties. This manual cleaning ensures the removal of contamination in the data; a caution we only exercise because of the Huygens landing site’s small area on the boundary of multiple terrain types.

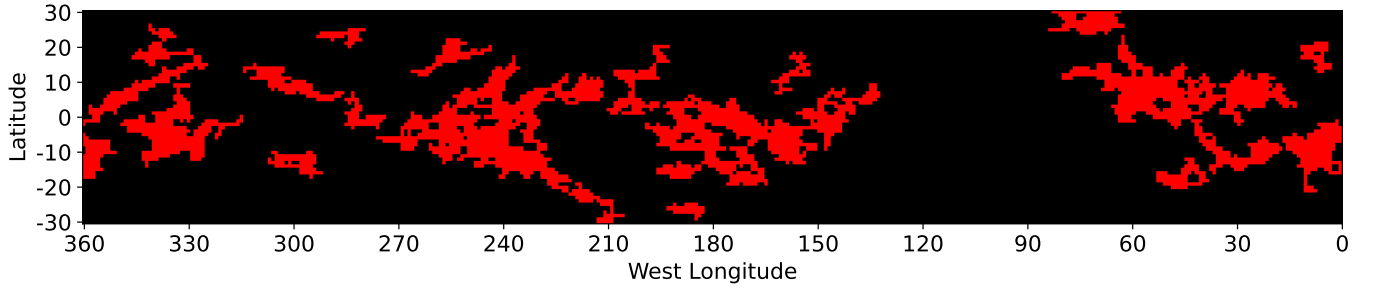
To counteract the lack of observational geometries for both the equatorial dunes and the Huygens landing site, we tetrahedrally interpolate to fill in as many observation angles as possible in the restricted BRDFs using the PyVista package (Sullivan & Kaszynski 2019).

## 5. SIMULATION VS DATA COMPARISON

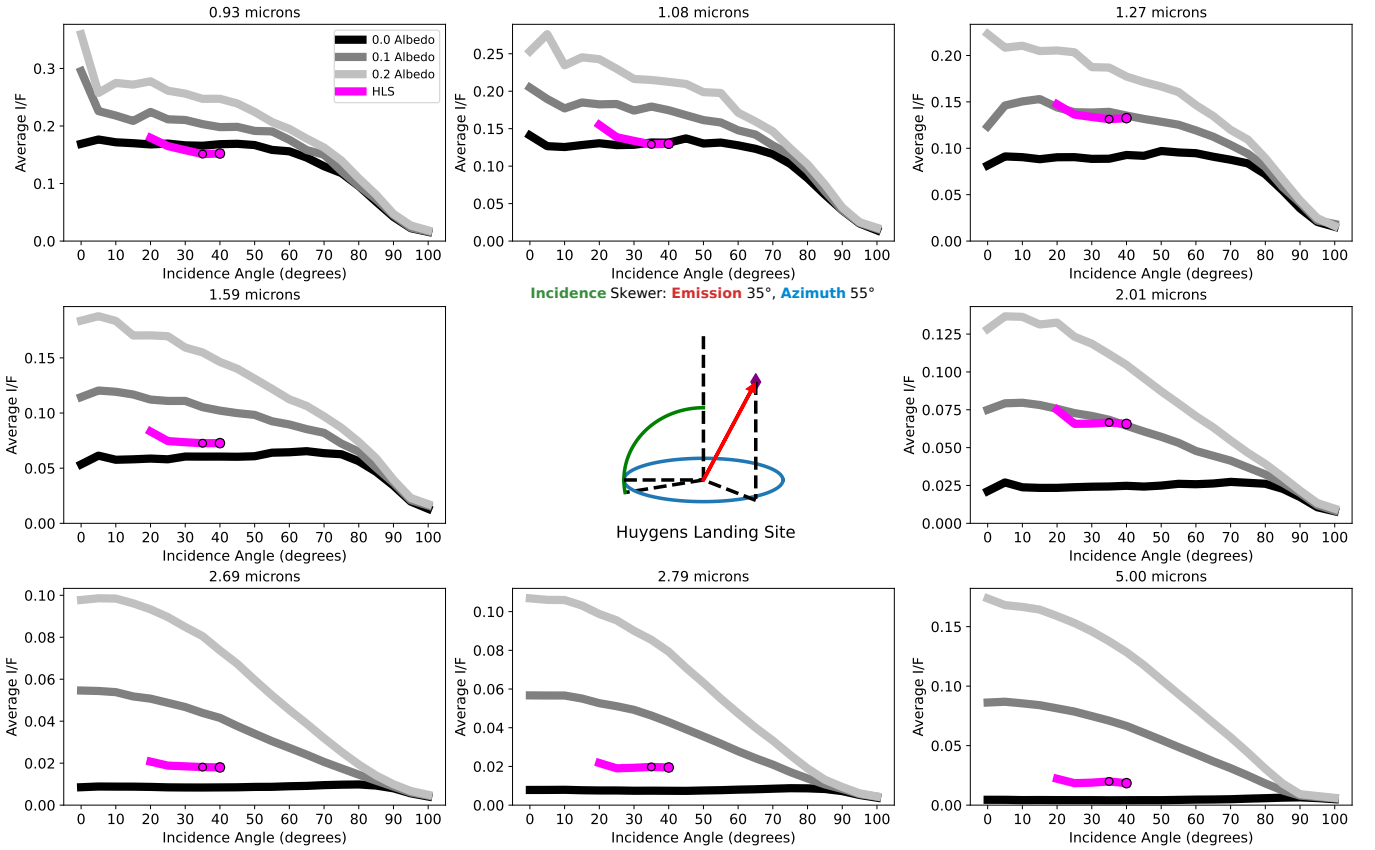
### 5.1. Huygens Landing Site

We choose to examine the Huygens landing site first due to its simpler data set, covering significantly fewer viewing geometries than the dunes. As three-dimensional data (incidence, emission, and azimuth angles, in this case) are hard to visualize, we opt to plot individual one-dimensional “skewers” of the restricted BRDFs at a time, fixing two of the three viewing geometry angles while allowing the third to vary. We create skewers for all of the restricted BRDFs, both simulations and observational data. We plot select representative skewers of the three simulations and Huygens landing site data in Figures 7-9. Each figure considers a different viewing angle along its x-axis: Figure 7 showcases the variations of I/F (radiance factor) as a function of incidence with emission fixed at angle at 35° and azimuth fixed at 55°, Figure 8 showcases the variations of I/F as a function of the emission angle with incidence fixed at 35° and azimuth fixed at 50°, and Figure 9 showcases the variations of I/F as a function of the azimuth angle with incidence fixed at 35° and emission fixed at 30°.

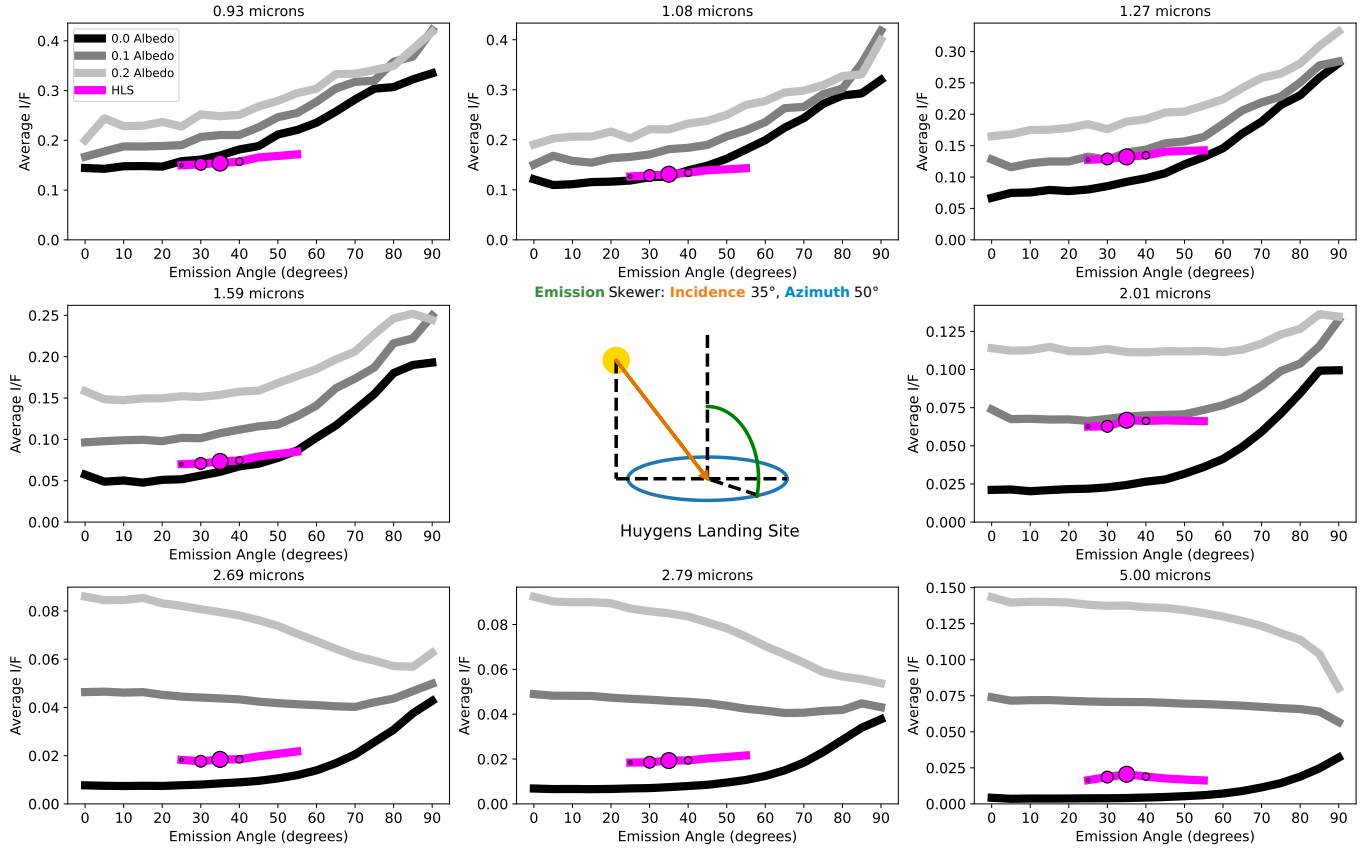
The simulations’ geometries naturally extend far beyond those actually acquired by Cassini. All three skewer types show a distinct shifting of behavior as a function of wavelength. In general, shorter wavelengths show higher I/F as



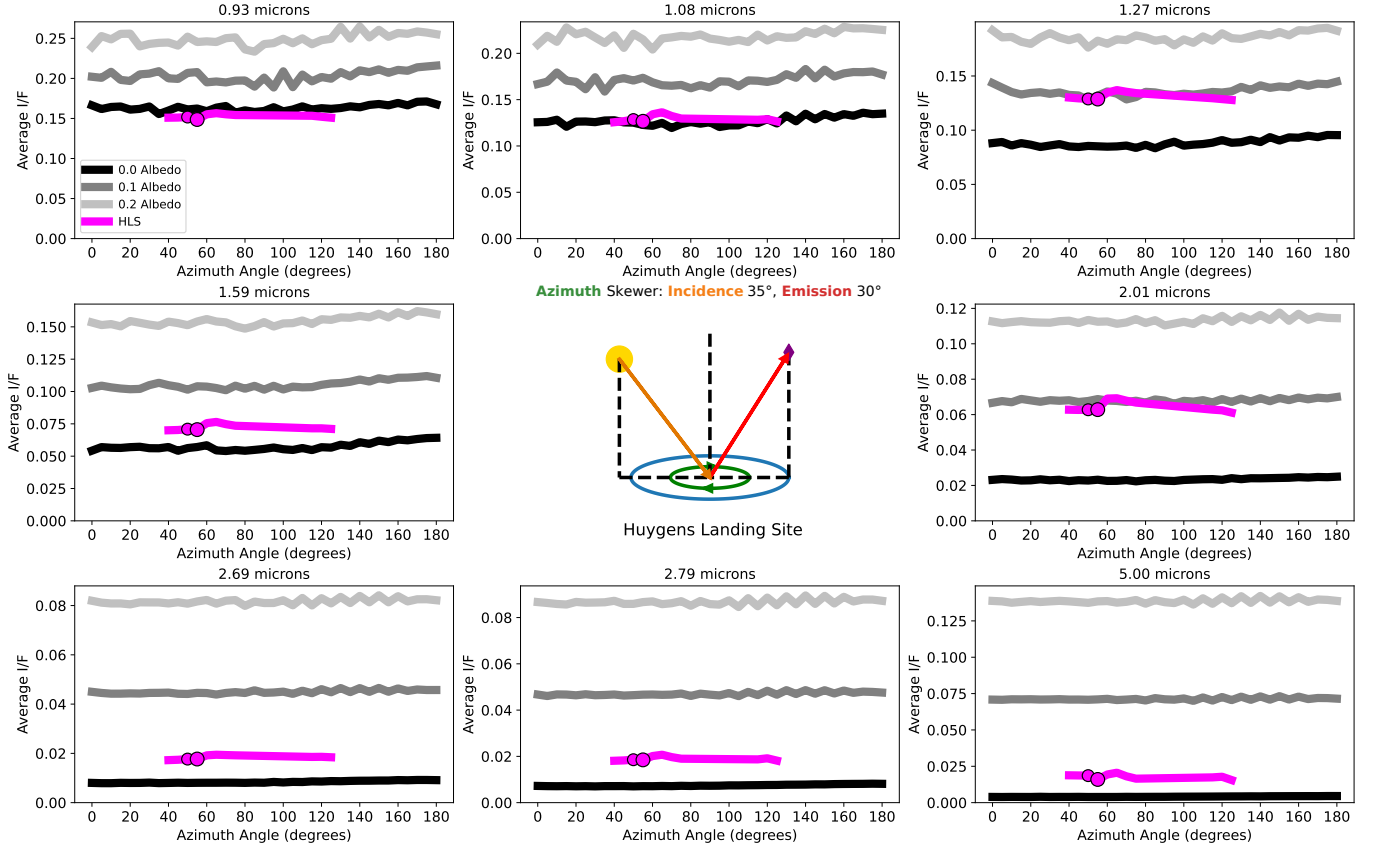
**Figure 6.** Titan equatorial surface terrain mask for dunes, as informed by [Lopes et al. \(2020\)](#) mixed with VIMS observations. The red pixels are dunes; black pixels are everything else. Almost all of the dunes on Titan exist in this mask, though a few small regions lie outside the  $30^\circ$  and  $-30^\circ$  latitude boundaries.



**Figure 7.** Average I/F (radiance factor) incidence angle skewers for the Huygens landing site with emission angle fixed at  $35^\circ$  and azimuth angle fixed at  $55^\circ$ . We arrange the eight wavelengths from shortest to longest starting from the top left corner. Lambertian simulation lines are monochromatic; the Huygens landing site data are pink. Bins with direct Huygens landing site observations have dots plotted over the lines, with larger dots indicating more observations. Bins on the Huygens landing site lines without dots are interpolated values. Note the differing vertical axis scales for each wavelength. The central area contains a geometry diagram illustrating the specific geometry bins considered for these skewers. A green arc represents all considered incidence angles, while a red vector coupled with a dotted line for perspective portrays the specific emission and azimuth relative to a blue azimuthal ring representing the surface. The Huygens landing site is broadly consistent with a Lambertian surface when examining the incidence angle.



**Figure 8.** Average I/F emission angle skewers for the Huygens landing site with incidence angle fixed at  $35^\circ$  and azimuth angle fixed at  $50^\circ$ . We arrange the eight wavelengths from shortest to longest starting from the top left corner. Lambertian simulation lines are monochromatic; the Huygens landing site data are pink. Bins with direct Huygens landing site observations have dots plotted over the lines, with larger dots indicating more observations. Bins on the Huygens landing site lines without dots are interpolated values. Note the differing vertical axis scales for each wavelength. The central area contains a geometry diagram illustrating the specific geometry bins considered for these skewers. A green arc represents all considered emission angles. We tilt the arc at the specific azimuth angle, using a vertical dotted line to give perspective. The orange line represents the incidence angle, or incoming light. The Huygens landing site is broadly consistent with a Lambertian surface when examining the emission angle, with the possible exception of 0.93 and 1.08  $\mu\text{m}$ , but those windows suffer from significant noise and unmodeled Rayleigh Scattering.



**Figure 9.** Average I/F azimuth angle skewers for the Huygens landing site with incidence angle fixed at  $35^\circ$  and emission angle fixed at  $30^\circ$ . We arrange the eight wavelengths from shortest to longest starting from the top left corner. Lambertian simulation lines are monochromatic; the Huygens landing site data are pink. Bins with direct Huygens landing site observations have dots plotted over the lines, with larger dots indicating more observations. Bins on the Huygens landing site lines without dots are interpolated values. Note the differing vertical axis scales for each wavelength. The central area contains a geometry diagram illustrating the specific geometry bins considered for these skewers. A green circle represents all considered azimuth angles, coinciding with the blue circle that represents the surface. The orange and red vectors depict the incidence and emission angles, respectively. The Huygens landing site is broadly consistent with a Lambertian surface when examining the azimuth angle.

light traverses through a higher optical depth of highly reflective haze aerosols (Es-sayeh et al. 2023), allowing for more scattering events in more directions. Longer wavelengths usually exhibit lower I/F, though  $5.00 \mu\text{m}$  is often slightly brighter than  $2.79 \mu\text{m}$ .

Incidence skewers almost always have a characteristic pattern, with the three different albedo simulations spread out in I/F at low incidence, lowering in I/F and coming closer together as incidence increases, until at the highest incidences the simulations become indistinguishable from each other. Figure 7’s skewers with emission fixed at  $35^\circ$  and with azimuth fixed at  $55^\circ$  exemplify the general incidence behavior.

Emission skewers follow a similar but inverted pattern to the incidence skewers and see more variability between the different windows. As a general rule, lower emission angles show the different albedo simulations spread out with relatively low I/F, and high emission has the simulations closer together with relatively high I/F. The shortest wavelengths

exhibit minimal difference between the simulations’ I/F, but I/F still rises with emission. The longer wavelengths start to deviate from the pattern where the 0.2 albedo simulation trends downward in I/F with higher emission, while the 0.1 albedo simulation lies flat, and only the 0.0 albedo simulation rises.  $5.00 \mu\text{m}$  almost always behaves this way. We attribute this behavior to atmospheric haze undergoing limb brightening at high emission, while at the same time surface features become limb darkened. Surface features have a greater influence on the clearer, longer-wavelength windows, hence the change in simulation behavior. The other long wavelengths change behavior based on the specific viewing geometry. Figure 8’s emission skewers with emission fixed at  $35^\circ$  and azimuth fixed at  $50^\circ$  exemplify this curious long wavelength behavior.

Azimuth skewers tend to be mostly flat in the simulation results, a behavior exemplified by Figure 9’s skewers at  $35^\circ$  incidence and  $30^\circ$  emission. Azimuth skewer behavior

changes when both incidence and emission approach larger values, as the forward-scattering haze begins to take effect.

Across these three figures, we display only a small selection of the total three-dimensional histogram; behavior can change significantly when skewering at different incidence, emission, or azimuth angles, though the transitions are always relatively smooth. We chose these particular skewers because they have the best Huygens landing site data.

The Huygens landing site data behaves well, for the most part, interpolating smoothly and forming lines that correspond rather well to the simulations. In 1.27 and 2.01  $\mu\text{m}$ , the data line up almost perfectly with the 0.1 albedo simulation. The 1.59, 2.69, 2.79 and 5  $\mu\text{m}$  windows have around 0.03 albedo.

The 1.08 and 0.93  $\mu\text{m}$  data hug the 0.0 albedo line from the simulations, and of all the wavelengths, they show the largest deviations from parallel to the simulation, most notable in Figure 8's emission skewers with incidence fixed at 35° and azimuth fixed at 50°, where the observational data dips below 0.0 albedo. We expected deviations for these two shortest of windows due to large atmospheric contributions, excess noise, and unmodeled Rayleigh scattering.

Even considering the deviations at short wavelengths, the Huygens landing site appears consistent with a Lambertian surface in the limited viewing geometries to which we have access. However, considering the relatively small section of the range these viewing geometries cover, we are skeptical of these results. We chose the best skewers we could for Figures 7-9, and even the azimuth skewers cover less than half of the available values. Cassini's observations of the Huygens landing site don't probe the higher incidence and emission angles, and we don't even approach the extremes. Thus, we move to a much larger data set: the equatorial dunes.

## 5.2. Equatorial Dunes

Similarly to the Huygens landing site data, we plot three representative one-dimensional skewer sets for the dunes, one for each of the observation dimensions. Each considers a different viewing angle along its x-axis: Figure 10 shows the variations of I/F as a function of incidence angle with emission fixed at 5° and azimuth fixed at 50°, Figure 11 showcases the variations of I/F as a function of emission angle with incidence fixed at 75° and azimuth fixed at 175°, and Figure 12 showcases the variations of I/F as a function of azimuth angle with incidence fixed at 40° and emission fixed at 30°.

In addition, since we have data covering a large range of viewing geometries, generating two-dimensional slices of the data becomes viable. In such plots, we still desire to make a direct comparison between reality and simulation, and so in Figure 13, we do not plot I/F, but instead retrieved surface albedo. All three of the geometries examined show a re-

trieved surface albedo for every emission and azimuth angle with valid observations at a single specified incidence angle and IR window wavelength. Specifically, we show a slice at fixed 55° incidence for 1.27  $\mu\text{m}$ , fixed 50° incidence for 2.01  $\mu\text{m}$ , and fixed 35° incidence for 5  $\mu\text{m}$ .

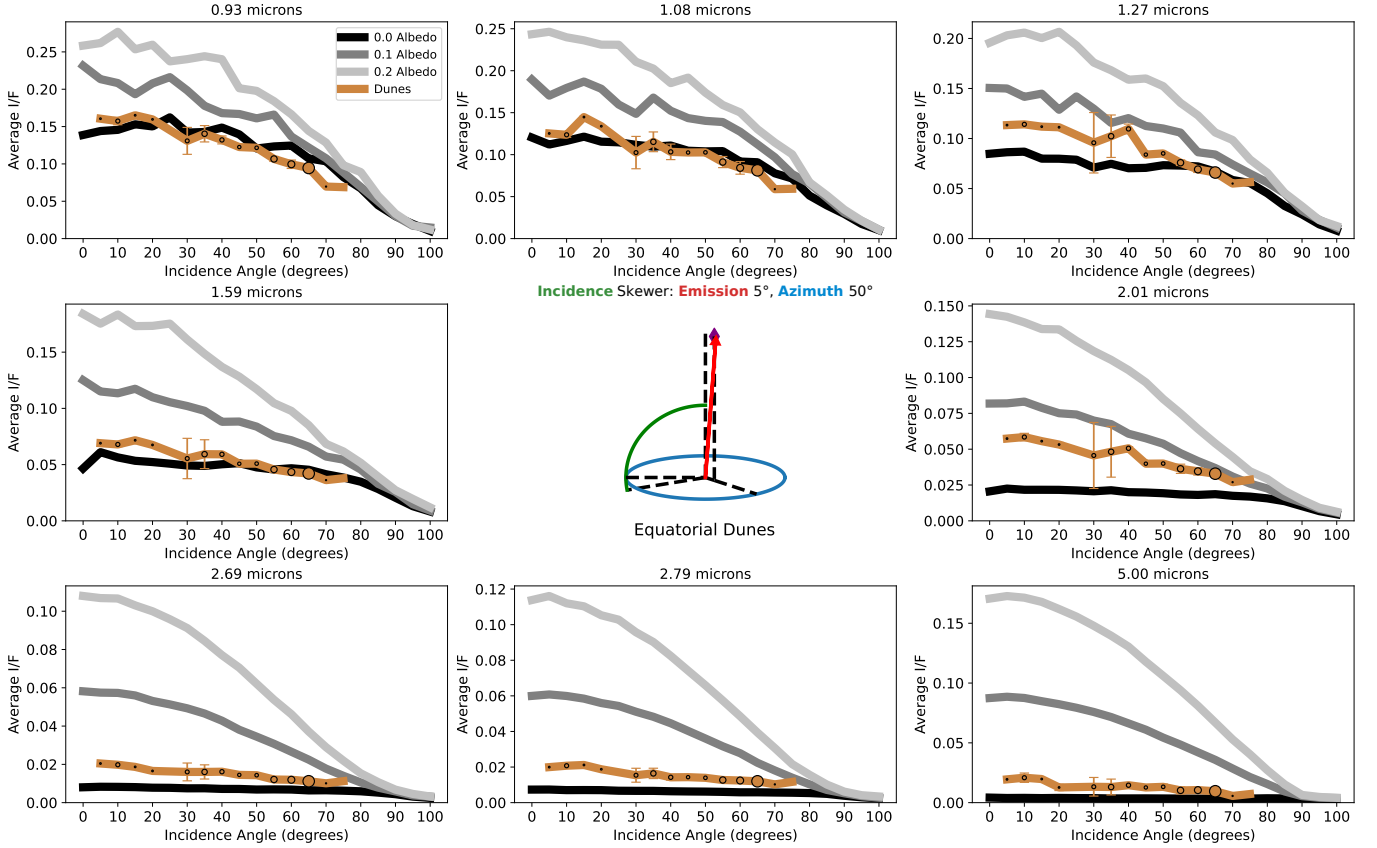
To retrieve the surface albedo, we make use of the three different albedos of the simulations: 0.0, 0.1, and 0.2. When the equatorial dunes data point lies between these points, we linearly interpolate to obtain an albedo value for the dunes' surface, and if the real data point is below the 0.0 albedo simulation or above the 0.2 albedo simulation, we extrapolate using a linear fit to retrieve the albedo. The fit sometimes creates unphysical negative retrieved albedos. We keep these negative values, so when plotted, some indication remains of how far below the predicted 0.0 albedo simulation the real dunes lie.

Examining the results, we find that the dunes data match the shape of the simulations rather closely in the incidence skewer views, as we can see in the skewers of Figure 10 with fixed 5° emission and fixed 50° azimuth. Incidence angles higher than 80° should be taken with a grain of salt, as Cassini acquired few observations near and beyond the terminator. When we examine the full set of data, we see some minor deviations in incidence skewers at high emission and high azimuth, but the limited observations in those geometries make it difficult to draw conclusions from the discrepancy, if the discrepancy truly exists.

The incidence skewers also clearly show a physical impossibility that sometimes crops up: multiple observations for 0.93 and 1.08  $\mu\text{m}$  dip below the 0.0 albedo simulation, resulting in a negative retrieved albedo. This retrieval occurs somewhat regularly within the shorter wavelength windows. In Figure 13, the slice taken with incidence fixed at 55°, shows quite a few negative retrieved albedos in the 1.27  $\mu\text{m}$  window. Considering that we predicted that the interdunes would brighten what we saw in the dunes observations, these overly dim retrievals come as a surprise.

Now, in some cases, the negative retrieved albedo occurs because the surface ceases to have a significant result on the simulations: at 100° incidence, the simulations do not show variation with albedo due to their atmospherically-dominated nature, forcing an unrealistic extrapolation. Such extrapolations happen at high emission angles as well, though to a lesser extent. Unfortunately, the unrealistic extrapolations do not explain every negative albedo, as the slice with fixed 55° incidence at 1.27  $\mu\text{m}$  in Figure 13 shows many negative albedos, and we have clear examples of observations falling below the dimmest simulation in many skewers.

The lack of Rayleigh scattering in SRTC++ may explain the physically impossible albedo retrievals. Rayleigh scattering is most relevant at short wavelengths (Es-sayeh et al. 2023), which commonly produce negative retrieved albedos.



**Figure 10.** Average I/F incidence angle skewers for the equatorial dunes with emission angle fixed at  $5^\circ$  and azimuth angle fixed at  $50^\circ$ . We arrange the eight wavelengths from shortest to longest starting from the top left corner. Lambertian simulation lines are monochromatic, and the equatorial dunes data are brown. Bins with direct dunes observations have dots plotted over the lines, with larger dots indicating more observations. Segments on the dunes lines without dots are interpolated values. We derive the error bars from each noninterpolated bin's standard deviation; most errors are too small to see. Note the differing vertical axis scales for each wavelength. The central area contains a geometry diagram illustrating the specific geometry bins considered for these skewers. A green arc represents all considered incidence angles, while a red vector coupled with a dotted line for perspective portrays the specific emission and azimuth relative to a blue azimuthal ring representing the surface. The equatorial dunes are broadly consistent with a Lambertian surface when examining the incidence angle in isolation.

Intuitively, one would expect adding Rayleigh scattering to make the simulation brighter, not dimmer, but subtle effects could be at play—to fully answer this question, we would have to implement Rayleigh scattering, which is on the list for future improvements to SRTC++.

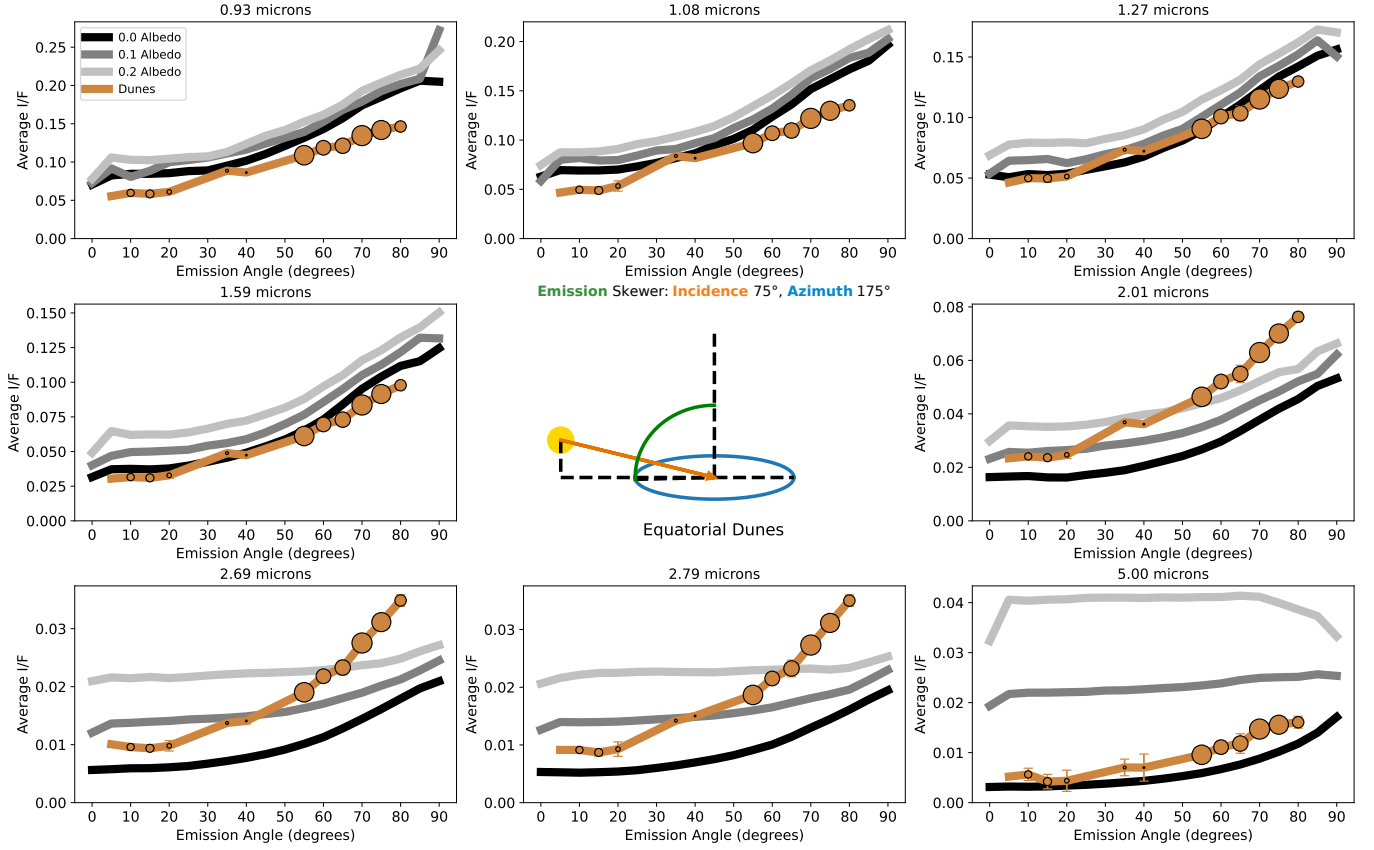
Notably, in the  $1.27 \mu\text{m}$ , fixed  $55^\circ$  incidence angle slice of Figure 13, the negative retrieved albedos cluster largely around the edge of the plot. Indeed, the plot as a whole shows a lowering of retrieved albedo with higher emission angle. We found this behavior in the data for the  $0.93$ ,  $1.08$ , and  $1.59 \mu\text{m}$  windows as well. This lowering of retrieved albedo with emission would not occur if the simulations accurately represented actual Titan. For a truly Lambertian surface, our procedure should retrieve relatively constant albedos, barring extreme situations where the simulations become nearly indistinguishable. Such a ideal cases appear in the  $5 \mu\text{m}$  albedo retrievals, including our selection at fixed  $35^\circ$  incidence in Figure 13. The  $5 \mu\text{m}$  window produces largely consistent

albedo retrievals; the only one of the eight windows to do so, potentially because of its lack of atmospheric influence.

If we examine the other remaining windows ( $2.01$ ,  $2.69$ , and  $2.79 \mu\text{m}$ ), we find results akin to those in the slice at  $2.01 \mu\text{m}$ , fixed  $50^\circ$  incidence in Figure 13, where the retrieved albedo increases at higher emission angles. Emission skewers such as the fixed  $75^\circ$  incidence and fixed  $175^\circ$  azimuth set in Figure 11 showcase this remarkably dramatic increasing of retrieved albedo with emission angle.

Up to around  $40^\circ$  emission, the dunes data match the simulations decently well, though the fixed  $75^\circ$  incidence and fixed  $175^\circ$  azimuth skewers in Figure 11 continue to demonstrate I/F values below the  $0.0$  surface albedo simulation at short wavelengths.

At emission angles past  $40^\circ$ , the simulations no longer broadly follow the observational data. The  $0.93 - 1.59 \mu\text{m}$  windows dip further below the  $0.0$  albedo simulation as emission increases, while  $2.01 - 2.79 \mu\text{m}$  do the opposite and



**Figure 11.** Average I/F emission angle skewers for the equatorial dunes with incidence angle fixed at  $75^\circ$  and azimuth angle fixed at  $175^\circ$ . We arrange the eight wavelengths from shortest to longest starting from the top left corner. Lambertian simulation lines are monochromatic, and the equatorial dunes data are brown. Bins with direct dunes observations have dots plotted over the lines, with larger dots indicating more observations. Segments on the dunes lines without dots are interpolated values. We derive the error bars from each noninterpolated bin's standard deviation; most errors are too small to see. Note the differing vertical axis scales for each wavelength. The central area contains a geometry diagram illustrating the specific geometry bins considered for these skewers. A green arc represents all considered emission angles. We tilt the arc at the specific azimuth angle, using a vertical dotted line to give perspective. The orange line represents the incidence angle, or incoming light. The equatorial dunes are notably inconsistent with a Lambertian surface when examining the emission angle, as evidenced by the dunes observations crossing the simulation lines of multiple different albedos. The notable exception of the  $5\mu\text{m}$  window remains broadly consistent with a Lambertian surface.

sharply tick above the simulations. As we can see in Figure 13 at  $1.27\mu\text{m}$ , fixed  $55^\circ$  incidence and at  $2.01\mu\text{m}$ , fixed  $50^\circ$  incidence, this effect is largely azimuthally independent, though the exact degree of the anomaly changes at forward or backscattering directions.

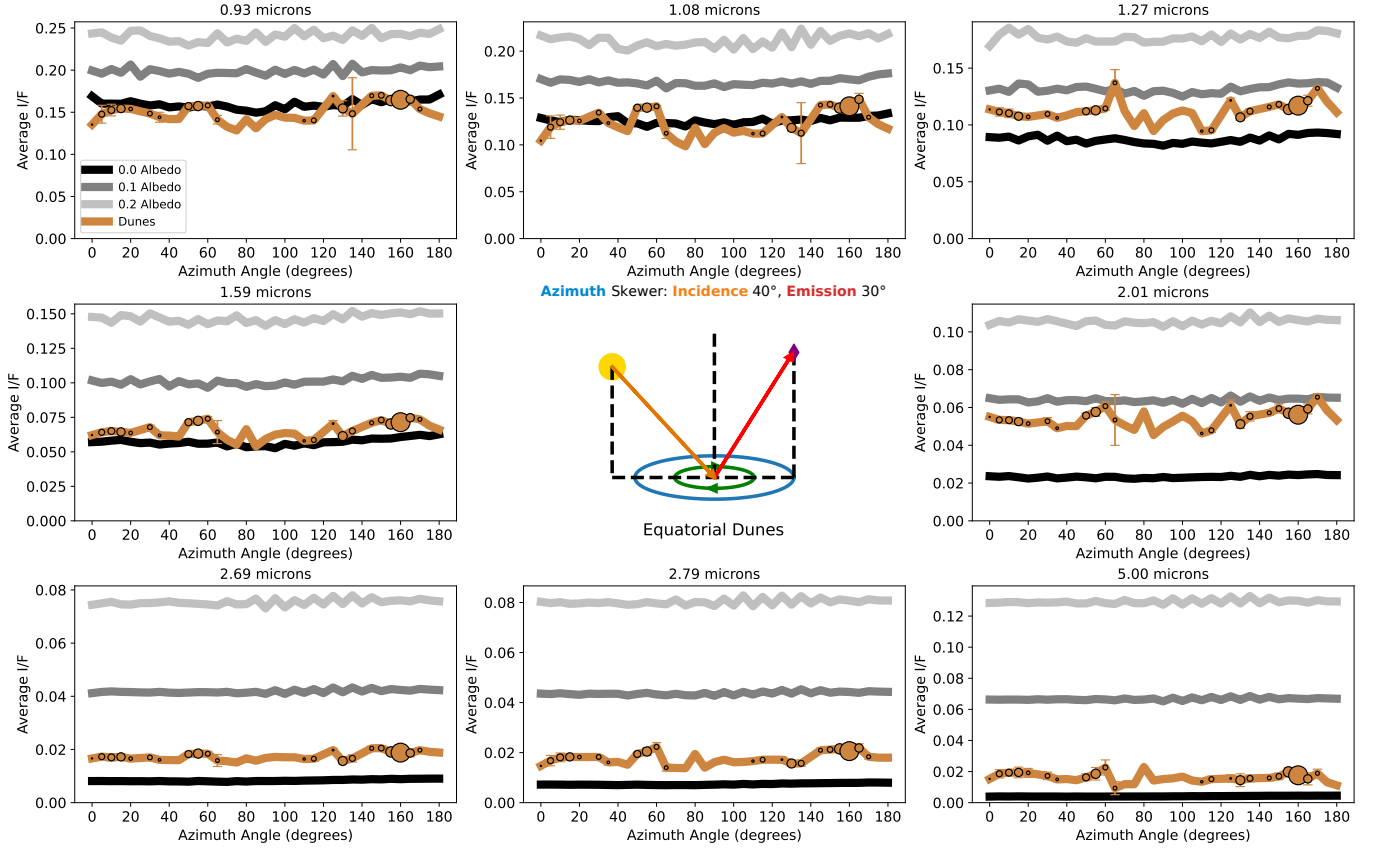
So, how can we explain the variability in retrieved albedo across the various emission angles? We identify three primary explanations: biased or contaminated data, real non-Lambertian surface properties, or inaccuracies in the atmospheric model.

We rule out biased or contaminated data rather quickly: the rise in retrieved surface albedo with emission exists in a wide variety of observations and viewing geometries, occurring at most azimuths. To check further, we divide up the restricted BRDFs based on the time of the observation, finding that the retrieved albedo still deviates with emission across the entire Cassini mission. Separating the data by latitude keeps

the effect, as does a longitudinal separation via specific dune fields. We could not find any variation in time or location on Titan. Thus, we rule out biased or contaminated data.

The second explanation for the rising retrieved albedo, a true non-Lambertian effect, seems unlikely. The brightening effect appears azimuthally agnostic: no matter what azimuth we consider, the emission effect exists where we have enough data to see it, and a forward-scattering or backscattering effect would focus at azimuths of  $0^\circ$  or  $180^\circ$ , respectively, not manifest everywhere. Topography-oriented explanations also seem unlikely, since the effect begins around  $40^\circ$  and only extreme emission angles should see influence by the muted topography that Cassini found (Corlies et al. 2017).

That said, we cannot entirely rule out the possibility of true non-Lambertian effects, for the dunes could behave in a completely unexpected manner. If we can rule out all other options, we would consider an unexpected surface prop-



**Figure 12.** Average I/F azimuth angle skewers for the equatorial dunes with incidence angle fixed at  $40^\circ$  and emission angle fixed at  $30^\circ$ . We arrange the eight wavelengths from shortest to longest starting from the top left corner. Lambertian simulation lines are monochromatic, and the equatorial dunes data are brown. Bins with direct dunes observations have dots plotted over the lines, with larger dots indicating more observations. Segments on the dunes lines without dots are interpolated values. We derive the error bars from each noninterpolated bin's standard deviation; most errors are too small to see. Note the differing vertical axis scales for each wavelength. The central area contains a geometry diagram illustrating the specific geometry bins considered for these skewers. A green circle represents all considered azimuth angles, coinciding with the blue circle that represents the surface. The orange and red vectors depict the incidence and emission angles, respectively. The equatorial dunes are broadly consistent with a Lambertian surface when examining the azimuth angle in isolation.

erty explanation in future work by testing out various non-Lambertian BRDFs for the surface in the `SRTC++` simulations and observing if any of them led to azimuthally agnostic effects.

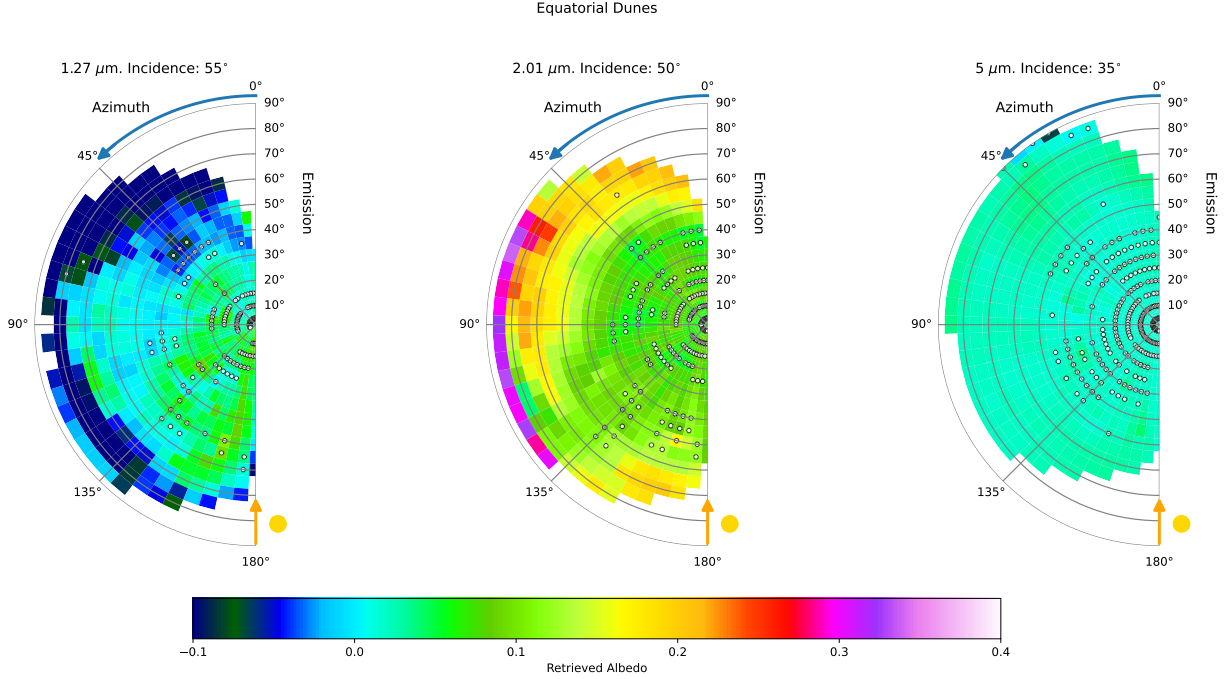
Lastly, we consider that the model may not accurately account for some atmospheric effect at high emission. While we mentioned Rayleigh scattering earlier, that will only have an important effect on short wavelengths (Es-sayeh et al. 2023). We identify three potential places in the atmospheric model where the deficiency could lie: the haze density, the atmospheric single scattering albedo (SSA), or the scattering phase function itself.

We examine the haze density and single scattering albedo scenarios by running simulations that modify those parameters. Figure 14 shows the results of altering the haze to 0.5 and 2.0 density, while adjusting the SSA by factors of 0.8 and 1.2. Since we only examine  $2\ \mu\text{m}$  here, which has SSA values of around 0.7, the factor of 1.2 cannot create unphysical

SSA values larger than 1. Unfortunately, none of the adjusted models come close to fixing the issue. The 0.5 haze simulation adds a strange hump at low emission, 2.0 haze exhibits minimal change, and 0.8 SSA flattens out the curve. The 1.2 SSA simulation actually shows a potentially correct slope at higher emission, but at lower emission, it fails completely!

These unsuccessful simulations lead us to the conclusion that the likely source of the discrepancy likely lies in the atmospheric scattering phase function. Unlike the other possibilities, simply increasing and decreasing some parameter would not sufficiently test the phase function, for it has a wide variety of shapes that it could take, and any adjustments to it would require normalization. Exploration of the influences of altering the phase function is beyond the scope of this paper, so we leave that investigation to future work.

Any of the three considered parameters (haze density, SSA, and phase function) could vary altitudinally in a way inaccurately represented in the model, which could also cause



**Figure 13.** Retrieved surface albedo slice for the 1.27, 2.01, and 5  $\mu\text{m}$  windows for the equatorial dunes at fixed incidence angles of 55°, 50°, and 35°, respectively. The orange arrow at the bottom of the figure indicates the direction of incoming sunlight. White points represent viewing geometries with real observations. Spaces with no white point represent interpolated values or, if white, missing values. We truncate retrieved albedos less than negative 0.1 to that value. These particular views showcase the overall lowering of retrieved albedo with emission for 1.27  $\mu\text{m}$ , the overall increase of retrieved albedo with emission for 2.01  $\mu\text{m}$ , and the relatively consistent retrieved albedo for 5  $\mu\text{m}$  of around 0.025.

the discrepancy. A combination effect from multiple parameters may also be involved. Investigating these possibilities is also left to future work.

Despite the clear deviation at high emission, overall, the simulation and dunes data still match remarkably well. The shape of the dunes data matches the incidence and azimuth skewer predictions, even if sometimes it sits below the 0.0 albedo simulation. Azimuth skewers in particular behave well, most exhibiting clearly flat behavior in the dunes data, such as in the skewers with incidence fixed at 40° and emission fixed at 30° in Figure 12.

With the shapes of the dunes data matching the restricted BRDFs so closely in incidence and azimuth, it seems reasonable to conclude that the dunes act substantially as a Lambertian surface. The primary evidence for non-Lambertian behavior comes from the inconsistent albedo retrieval at variable emission, which does not appear to have an azimuthal dependence, making it likely an atmospheric effect not accounted for in the simulation, rather than a true non-Lambertian effect.

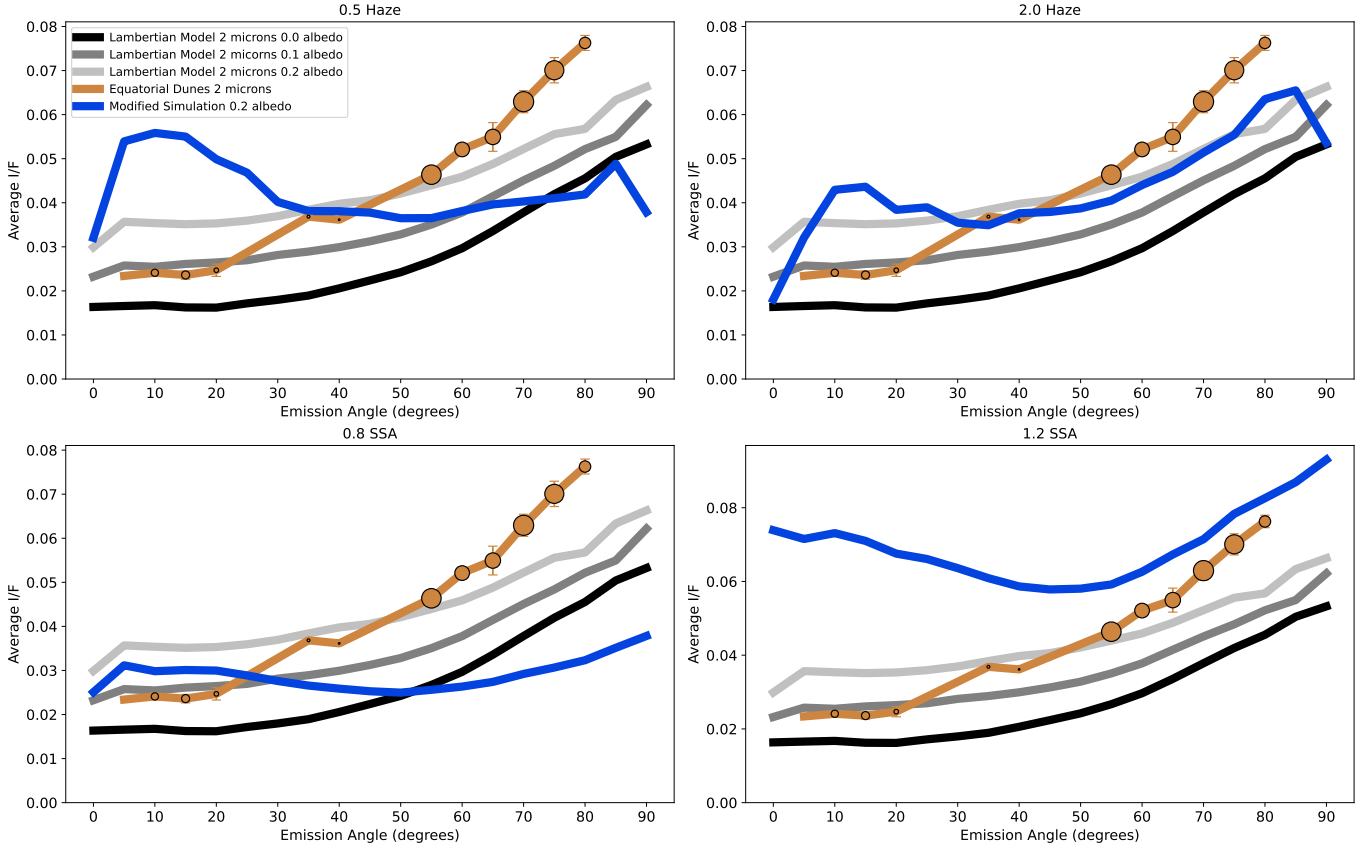
### 5.3. Retrieved Surface Albedos

As a final, more quantitative check on validation, we compare our retrieved surface albedos with those reported for the dunes in prior literature (McCord et al. 2006; Bonnefoy et al. 2016; Solomonidou et al. 2018; Brossier et al. 2018;

Es-sayeh et al. 2023). In order to come up with a singular retrieved albedo value, we consider bins that contain real observations, ignoring interpolated values. We sort each bin by wavelength and then combine them into a single data set. As we have reason to suspect our methods less reliable at extreme viewing geometries, we limit our considered observations to incidence and emission angles of 40° or less, as we identified the emission angle discrepancy to start at 40°. Figure 15 contains the results of the surface albedo retrievals.

We select the median of the data as our overall retrieved albedo and obtain uncertainties from the median absolute deviation (MAD) times 1.4826. We present the numerical values of the medians and modified MAD values in Table 1, including two sets of values: those restricted to incidence and emission angles less than 40°, and those calculated with all observed values. In both Table 1 and Figure 15 we include dunes albedos retrieved from prior work in the literature.

Our albedo retrieval for 5  $\mu\text{m}$  demonstrates consistency with all previous work. The double windows 2.69 and 2.79  $\mu\text{m}$  show good agreement as well, though already at this point we can see prior work diverging into two separate camps: McCord et al. (2006), Es-sayeh et al. (2023), and our work on one side; Bonnefoy et al. (2016), Lopes et al. (2016), and Brossier et al. (2018) on the other. Solomonidou et al. (2018) reports lower albedo values broadly consistent with ours when they consider multiple locations on Titan, but



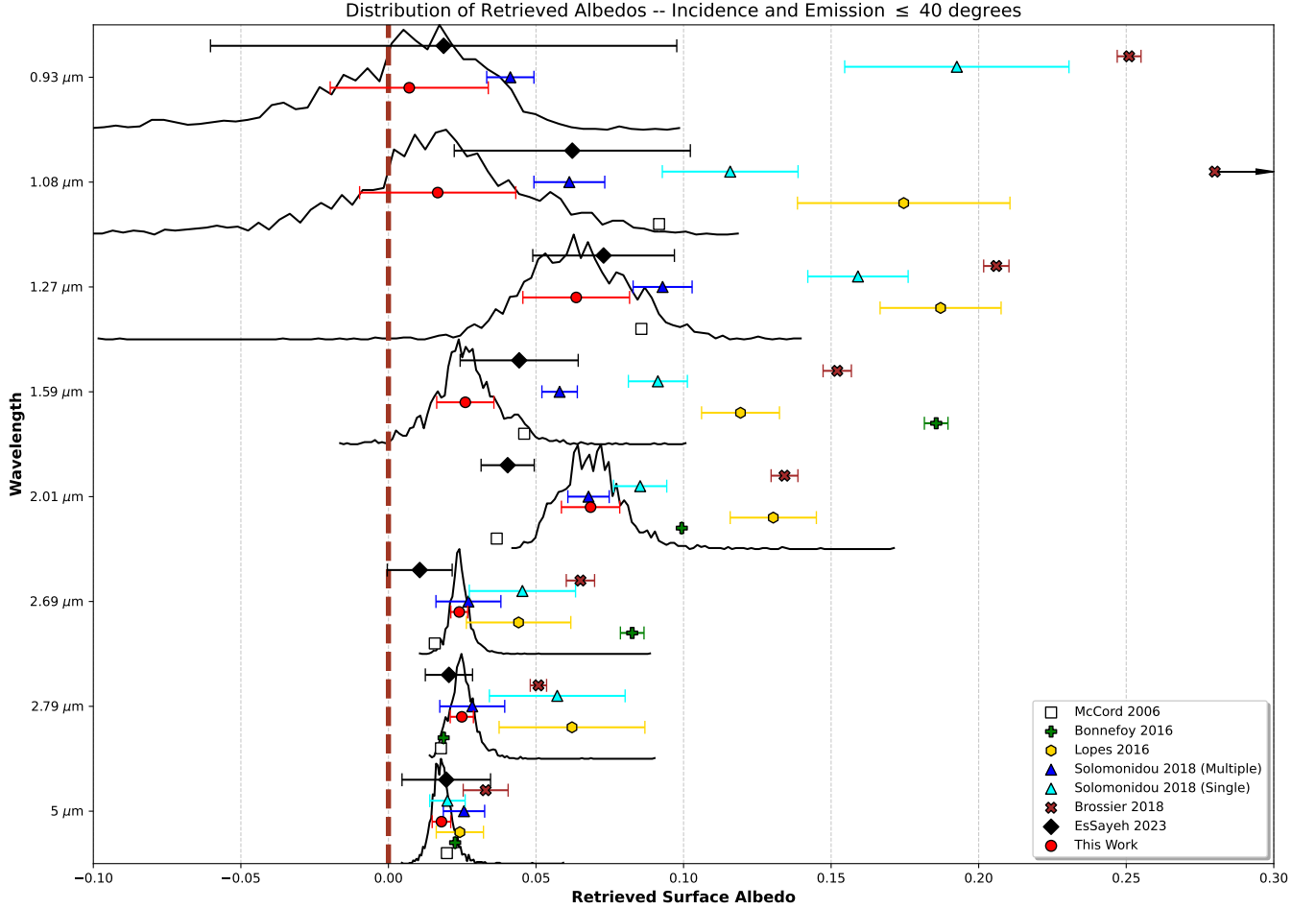
**Figure 14.** Emission skewer showcases of altered haze density and atmospheric single scattering albedo (SSA) simulations at  $2\ \mu\text{m}$  with fixed  $75^\circ$  incidence angle and fixed  $175^\circ$  azimuth angle. This figure shares its viewing geometry with that of Figure 11. The monochromatic lines show the unaltered simulations at various albedos, while the brown line contains the real dunes data. We make error bars from each bin's standard deviation for noninterpolated bins. The monochromatic and brown lines are identical in all four displays; the blue lines differ and represent simulations run at 0.2 surface albedo with some atmospheric property adjusted. The top left is 0.5 haze density, the top right is double the haze density. The bottom left is SSA dimmed by a factor of 0.8, while the bottom right is SSA brightened by 1.2. Notably, none of the adjustments produce the kind of change that would scale to match the shape of the real dunes data on its own.

they also report values for a single location on Titan more consistent with the brighter albedo values of Bonnefoy et al. (2016), Lopes et al. (2016), and Brossier et al. (2018).

We are unsure why we see such drastic differences between these retrievals. The different results in Solomonidou et al. (2018) suggest that only looking at a single place on Titan may bias the data. This interpretation is complicated by Bonnefoy et al. (2016), which averages several different locations, retrieving albedos consistent with the brighter results, not the dimmer ones, which are where our result and Solomonidou et al. (2018)'s multiple locations result reside. Notably, Bonnefoy et al. (2016) and Solomonidou et al. (2018) cover far less area than our survey. While unlikely, the possibility exists that one or both surveys sampled a biased subset of Titan's dunes that had different albedo responses. We also have to consider bias in our data: while our results are global in nature, the restrictive nature of our mask ensures we never sample dunes anywhere near another terrain type, which can create bias if central dunes differ from the edges.

Model differences could easily cause such incongruities in retrieved albedo. However, Lopes et al. (2016); Solomonidou et al. (2018); Brossier et al. (2018) and Es-sayeh et al. (2023) all originate from the same Hirtzig et al. (2013) model, yet these four produce both high-albedo and low-albedo retrievals. Lopes et al. (2016) and Solomonidou et al. (2018) appear to use near identical procedures to each other, and while some agreement exists between Solomonidou et al. (2018)'s single location results and Lopes et al. (2016)'s results, the agreement is inconsistent.

Leaving aside the brighter albedos for a moment, our results show broad agreement with the dimmer retrieved albedo values; even our retrievals for the  $0.93\ \mu\text{m}$  and  $1.08\ \mu\text{m}$  windows, which we identify as having poor correlation with reality considering that their distributions dip significantly into negative retrieved albedos. Given the large error bars placed on measurements in this short wavelength regime, when retrieved albedos are even reported, we suggest that research



**Figure 15.** Retrieved surface albedo distribution plots compared with retrieved surface albedos of prior studies. These plots only consider retrieved albedos at incidence and emission angles lower than  $40^\circ$ . The black lines show the distribution of retrieved albedos; we sorted each wavelength’s retrieved albedos into 100 bins and then scaled the resulting distribution so the most prevalent bin held a value of 1. The dark red dashed line marks 0.0 retrieved albedo, all retrieved values less than this are unphysical. The red points indicate the medians of our retrieved surface albedos, while our error bars are median absolute deviations (MAD) times 1.4826. We derived other error bars from the report of the indicated source. [McCord et al. \(2006\)](#) reported no uncertainties or error bars; all other points from previous work have them, though in many cases the errors are vanishingly small. [Brossier et al. \(2018\)](#) reports their values as surface single scattering albedos, which we convert to Lambertian surface albedos via procedures outlined in [Hapke \(2012\)](#). [Brossier et al. \(2018\)](#) provided a single point at  $1.08 \mu\text{m}$  that sits beyond 0.3 retrieved surface albedo. [Solomonidou et al. \(2018\)](#) reports two retrieved albedo sets, one for a single location on Titan, and another considering data from multiple locations. If no point exists in a window for a source, then that source did not report an albedo value for that window.

should not use the  $0.93$  and  $1.08 \mu\text{m}$  windows to draw conclusions about Titan’s surface in isolation.

Upon noting the relative agreement our retrieved albedos have with three different sources, our simulation results appear reasonable. Though, noting the wide spread of results across multiple studies, we conclude that the community’s knowledge of Titan’s surface albedo is limited and likely to change as new techniques arise.

#### 5.4. Opposition Effect

We recognize that our method of binning spectels together and averaging them all makes us significantly less sensitive to dramatic changes over tiny angular extents, such as the sharp

central peak often seen in the opposition effect, sometimes less than one degree away from opposition ([Kulyk 2008](#); [Schaefer et al. 2008](#)).

We perform a direct search across our entire database, image by image, for an opposition effect in the dunes (or an equivalent effect in the forward-scattering direction). We find no evidence for any such effect. We expect this for seven of the eight windows as diffuse lighting via scattering all but removes the opposition effect ([Schröder & Keller 2008](#)). However, the eighth window,  $5 \mu\text{m}$ , sees significant unimpeded sunlight even at sunset ([Barnes et al. 2018a](#)). Even here, we find no evidence of an opposition effect.

| Wavelength         | Retrieved Albedos   |                     |        |                   |                   |                   |                   |                   |
|--------------------|---------------------|---------------------|--------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                    | Full                | Truncated           | McCord | Bonnefoy          | Lopes             | Solomonidou       | Brossier          | Es-sayeh          |
|                    | Retrieved<br>Albedo | Retrieved<br>Albedo |        |                   |                   |                   |                   |                   |
|                    |                     |                     | 2006   | 2016              | 2016              | 2018              | 2018              | 2023              |
| 0.93 $\mu\text{m}$ | $-0.016 \pm 0.05$   | $0.007 \pm 0.03$    | -      | -                 | -                 | $0.041 \pm 0.008$ | $0.251 \pm 0.004$ | $0.019 \pm 0.07$  |
| 1.08 $\mu\text{m}$ | $0.002 \pm 0.03$    | $0.017 \pm 0.02$    | 0.092  | -                 | $0.175 \pm 0.04$  | $0.061 \pm 0.012$ | $0.370 \pm 0.003$ | $0.062 \pm 0.07$  |
| 1.27 $\mu\text{m}$ | $0.054 \pm 0.019$   | $0.064 \pm 0.012$   | 0.086  | -                 | $0.187 \pm 0.02$  | $0.093 \pm 0.01$  | $0.206 \pm 0.004$ | $0.073 \pm 0.02$  |
| 1.59 $\mu\text{m}$ | $0.02 \pm 0.012$    | $0.026 \pm 0.006$   | 0.046  | $0.186 \pm 0.004$ | $0.119 \pm 0.13$  | $0.058 \pm 0.006$ | $0.152 \pm 0.005$ | $0.044 \pm 0.02$  |
| 2.01 $\mu\text{m}$ | $0.071 \pm 0.009$   | $0.068 \pm 0.007$   | 0.037  | $0.099 \pm 0$     | $0.13 \pm 0.015$  | $0.068 \pm 0.007$ | $0.134 \pm 0.005$ | $0.040 \pm 0.009$ |
| 2.69 $\mu\text{m}$ | $0.025 \pm 0.003$   | $0.024 \pm 0.002$   | 0.016  | $0.083 \pm 0.004$ | $0.044 \pm 0.017$ | $0.027 \pm 0.011$ | $0.065 \pm 0.004$ | $0.011 \pm 0.011$ |
| 2.79 $\mu\text{m}$ | $0.027 \pm 0.004$   | $0.025 \pm 0.003$   | 0.018  | $0.019 \pm 0$     | $0.062 \pm 0.02$  | $0.028 \pm 0.007$ | $0.051 \pm 0.003$ | $0.021 \pm 0.008$ |
| 5 $\mu\text{m}$    | $0.018 \pm 0.004$   | $0.018 \pm 0.003$   | 0.02   | $0.023 \pm 0$     | $0.024 \pm 0.008$ | $0.026 \pm 0.007$ | $0.033 \pm 0.008$ | $0.02 \pm 0.015$  |

**Table 1.** Table of retrieved albedos for all eight IR wavelength windows, including albedos reported in other publications. The "full" retrieved albedos consider real observations at all angles, the "truncated" retrieved albedos only consider incidence and emission angles of  $40^\circ$  or less. We report the medians as the albedo values obtained for this paper, with uncertainties calculated by the median absolute deviation (MAD) times 1.4826. The values and errors obtained for other publications were taken as shown in the original publications, taken directly from the plots. Any empty entries were not reported or were not retrievable. [McCord et al. \(2006\)](#) did not report errors. The zero errors from [Bonnefoy et al. \(2016\)](#) indicate errors that were too small to see on the paper's plots. The reported values from [Solomonidou et al. \(2018\)](#) only include the case that considers multiple locations on Titan, rather than the singular location values.

To be fair, no observation we consider gets closer to opposition than between 1 and 2 degrees. Thus, our observations do not rule out a narrow and sharp opposition effect, as confinement to within  $0.1^\circ$  can occur. ([Kulyk 2008](#); [Schaefer et al. 2008](#)). The solar disc's footprint on Titan sits around  $0.05$  deg ([Barnes et al. 2011](#)), further implying undetectability of the opposition effect. Huygens observed the opposition effect spike to begin around  $0.2^\circ$  ([Karkoschka et al. 2012](#)), and that observation was made with Huygens' own illumination, not the Sun's. Furthermore, Huygens did not land on the dunes.

The viewing geometry to definitively confirm or deny the opposition effect does not exist in the Cassini VIMS data for the dunes; we can only say that if the dunes have an opposition effect, it is a narrow one with little-to-no broad component.

We also find no pseudo-specular forward-scattering component in the dunes, even though we did find an observation that approached zero degrees from the specular direction.

## 6. SUMMARY AND CONCLUSION

We have two primary purposes to this paper: to constrain Titan's surface dune properties and to validate the SRTC++ simulations against real data.

On the validation front, SRTC++ in general calculates restricted BRDF trends that match the real data in incidence and azimuth angles, but sometimes produces inconsistent albedos, as evidenced by situations where retrieve impossible albedos below 0.0. As these moments primarily occur at short wavelengths, the lack of modeled Rayleigh scattering could produce these impossible albedos. Alternatively, or perhaps additionally, the negative albedos could arise from

the faulty characterization of atmospheric haze—the far more prevalent observed deviations with increasing emission angle imply as much.

More quantitatively, our general retrieved albedos for the dunes in each of the individual windows are largely consistent with previous work. Our inquiry as to why two different trends in retrieved surface albedos for the dunes exist across the literature was inconclusive. Our results corroborate the lower albedo values rather than the higher albedo values.

Despite these caveats, SRTC++ still produces smooth lines parallel to the real data in most incidence and azimuth skewers, only with inconsistent surface albedo retrieval. As such, the simulations remain useful for characterizing Titan's surface, particularly when looking for clear deviations from the Lambertian assumption. We found almost no such deviations. We also found no evidence of an opposition effect and no evidence for a forward scattering component to the surface BRDF. The only major deviation we found is the dimming and brightening some windows exhibit at high emission angles—which is likely a haze model insufficiency rather than a real effect. The Huygens landing site also shows Lambertian behavior, though we do not say this with confidence considering the limited nature of the observations at that location.

We ultimately conclude that Titan's dunes are Lambertian surfaces, or very close to it. We expected this result, given the generally Lambertian nature of Earth's sand ([Hapke & van Hoen 1963](#)), though it exhibits an opposition effect within two degrees of zero phase angle ([Wise & Mars 2022](#)), which we would have seen if such an effect present on Titan. However, the width of an opposition effect likely narrows with distance from the Sun ([Karkoschka et al. 2012](#)) due to

the smaller angular diameter of the solar disk, so we cannot say that we should have seen an opposition effect on the dunes. Furthermore, characteristic differences exist between the dunes of Earth and Titan. Earth’s deserts tend toward high albedos, while Titan’s dunes exhibit the lowest albedo classification on Titan’s solid surface.

Future work would include improving SRTC++’s model of Titan’s haze to simulate the atmosphere properly at high emission—or, if that were to prove impossible, investigating what other effects might cause the inaccuracies.

This work could also assist in analysis of the specular lakes at Titan’s north pole; while the atmosphere is not as well characterized there, we can use the straightforward specular lake surface phase functions to probe where exactly the most poorly characterized parameters are. We can also examine other terrain types on Titan—though the investigation would

require a more manual rather than automated approach, as the other terrains often blend into each other without sharp distinctions.

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